

Consider a random experiment of rolling a fair die twice. If we have bet ₹ 100 on the sum of the outcomes being 5, then we are not really interested in how we get the sum 5 (i.e., the outcomes being (1,4) or (2,3) don't make any difference). All we care about is that whether the sum is 5 or not.

This is the motivation to come up with the concept of random variables.

Definition [Real-valued random variables]:

A real-valued random variable or simply random variable is a function from the sample space to the set of real numbers.

Examples:

1.

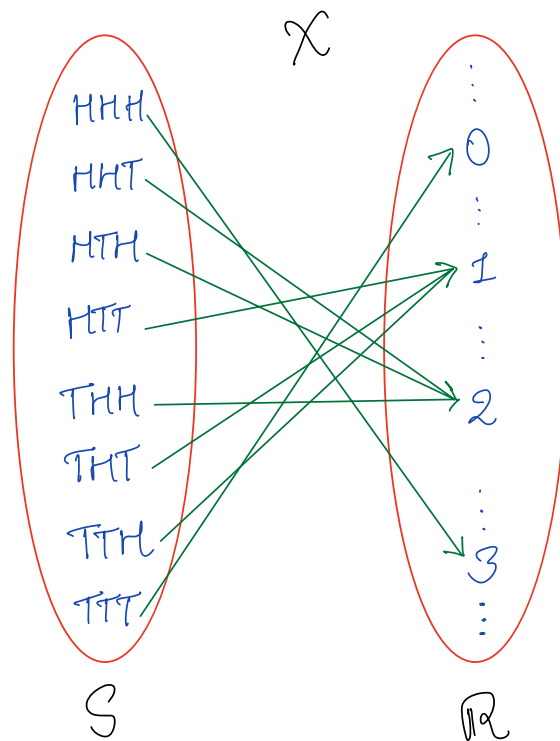
Suppose the random experiment is tossing a fair coin thrice.

Then the sample space S is,

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Then we can define following random variable X ,

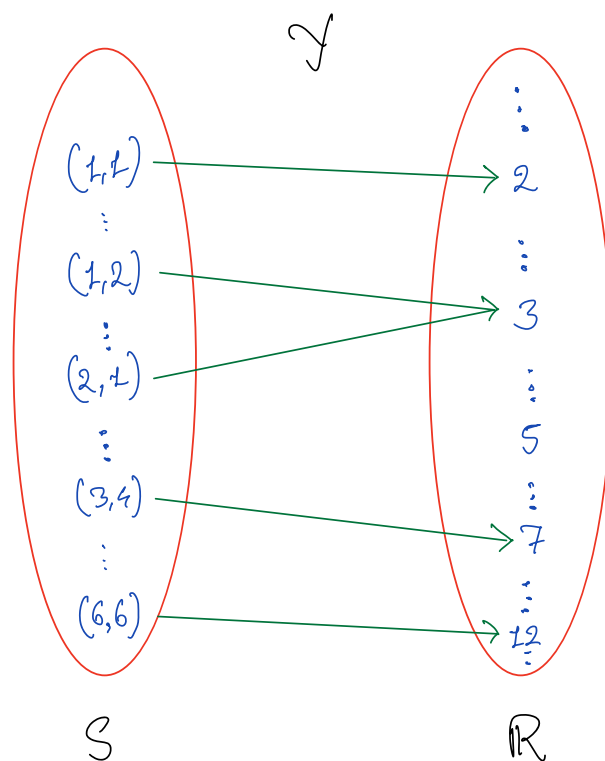
$$X: S \rightarrow \mathbb{R}$$



As we can observe, the random variable X tells us the number of the heads in our outcome.

2. As for another example, consider the experiment of rolling a fair die twice. Then we can define a random variable Y that corresponds to the sum of the 2 outcomes.

So,



Discrete Random Variable:

A random variable is said to be discrete if the range of the random variable is a countable set.

Both the examples 1. and 2. given above are examples of a discrete random variables where the range set is, $\{0, 1, 2, 3\}$ and $\{2, 3, \dots, 12\}$ respectively.

Definition [Probability Mass Function (PMF)]:

for a discrete random variable X ,
the probability mass function, denoted
by p is defined as,

$$p: \mathbb{R} \rightarrow \mathbb{R}$$

$$\text{s.t.} \quad \begin{aligned} p(a) &= \Pr\{X=a\} & , \text{ if } a \in \text{Range}(X) \\ p(a) &= 0 & , \text{ otherwise} \end{aligned}$$

So for example 1. given above,

We have,

$$p(0) = \Pr\{X=0\} = \frac{1}{8}$$

$$p(1) = \Pr\{X=1\} = \frac{3}{8}$$

$$p(2) = \Pr\{X=2\} = \frac{3}{8}$$

$$p(3) = \Pr\{X=3\} = \frac{1}{8}$$

$$p(a) = 0 \quad , \text{ if } a \notin \{0, 1, 2, 3\}$$

Similarly for example 2. given above,
We have,

$$p(2) = \Pr\{Y=2\} = 1/36$$

$$p(3) = \Pr\{Y=3\} = 2/36$$

$\vdots$

$$p(a) = 0, \text{ if } a \notin \{2, 3, \dots, 12\}$$

In general, if a discrete random variable X takes values $x_1, x_2, x_3, \dots$, then

$$p(x_i) = \Pr\{X=x_i\} \geq 0, \text{ for } i=1, 2, 3, \dots$$

and

$$p(x) = 0, \text{ if } x \notin \text{Range}(X)$$

Moreover,

$$\sum_{i=1}^{\infty} p(x_i) = 1$$

Expected value of a (discrete) random variable :

Suppose X is a discrete random variable that takes values $x_1, x_2, x_3, \dots$.

Then expected value of X denoted as,

$$E[X] = \sum_{i=1}^{\infty} x_i \cdot p(x_i)$$

where p is the PMF of X .

Eg.

for random variable X defined in Eg. 1. above,

$$\begin{aligned} E[X] &= 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} \\ &= \frac{12}{8} = 1.5 \end{aligned}$$

Expectation of a function of a random variable.

If X is a random variable that takes values $x_1, x_2, x_3, \dots$ and if g is a real valued function, i.e.,

$$g: \mathbb{R} \rightarrow \mathbb{R}$$

Then $g(X)$ or $g \circ X$ is again a random variable and,

$$E[g(X)] = \sum_{i=1}^{\infty} g(x_i) \cdot p(x_i)$$

where p is the PMF of X .

Example

Consider the random variable X defined in example 1. above.

Suppose, $g: \mathbb{R} \rightarrow \mathbb{R}$

$$\text{s.t. } g(x) = x^2.$$

Then $g(X)$ or $g \circ X$ is yet again a random variable.

Since the $\text{Range}(X) = \{0, 1, 2, 3\}$

So,

$$E[g(X)] = g(0) \cdot P\{X=0\} + g(1) \cdot P\{X=1\} \\ + g(2) \cdot P\{X=2\} + g(3) \cdot P\{X=3\}$$

$$= 0^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{3}{8} + 2^2 \cdot \frac{3}{8} + 3^2 \cdot \frac{1}{8}$$

$$= \frac{24}{8} = 3$$

In general, since the function g above is defined as $g(x) = x^2$, we denote the random variable $g(X)$ as X^2 . If the function g was defined as, $g(x) = 2x + 3$, then the random variable $g(X)$ is denoted as $2X + 3$.

Variance of a random variable.

If X is a random variable that takes values $x_1, x_2, x_3, \dots$.

Then, variance of X is defined as,

$$\text{Var}(X) = E[(X - \mu)^2]$$

$$\text{where, } \mu = E[X]$$

We also have,

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Example.

Again, for the random variable X defined above in Example 1.

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

We have calculated $E[X^2]$ & $E[X]$ above,

$$\text{So, } \text{Var}(X) = 3 - (1.5)^2 = 0.75$$

Standard deviation of a random variable.

for a random variable X , its standard deviation denoted as $SD(X)$ is,

$$SD(X) = \sqrt{\text{Var}(X)}$$